

C1 extends C2, C3

Program Code:

```
package core_java;

class C3 {
    void openLibrary() {
        System.out.println("Library is open");
    }
}

class C2 extends C3 {
    void issueBook() {
        System.out.println("Book issued successfully");
    }
}

class C1 extends C2 {
    void readBook() {
        System.out.println("Reading a Java book");
    }
}

public class C1toC2C3 {
    public static void main(String[] args) {
        C1 obj = new C1();

        obj.openLibrary();
        obj.issueBook();
        obj.readBook();
    }
}
```

Explanation:

This program demonstrates multilevel inheritance using the classes C1, C2 and C3. C1 extends C2 and C2 extends C3. Therefore, an object of C1 can access methods defined in C1, C2 and C3.

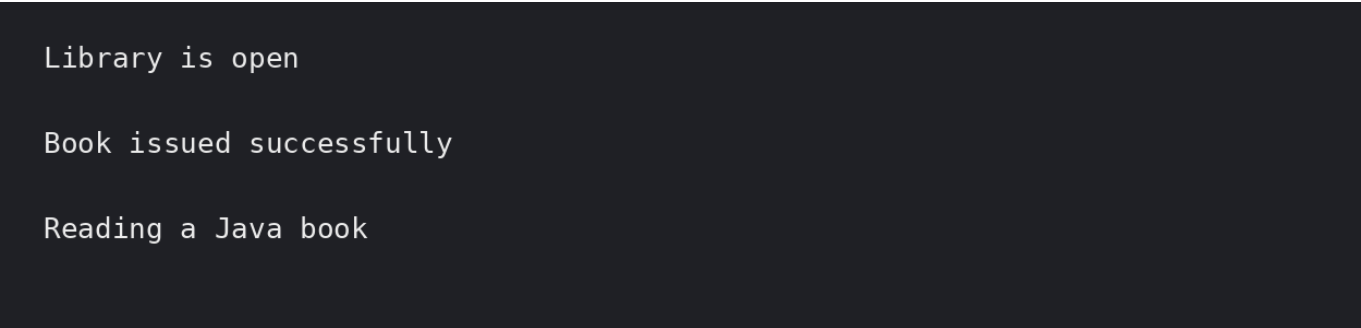
Compilation and Execution Commands:

Save the file as: C1toC2C3.java

Compile:
javac -d . C1toC2C3.java

Run:
java core_java.C1toC2C3

Output Screenshot:



```
Library is open

Book issued successfully

Reading a Java book
```

Verification:

The output confirms that the C1 object successfully accesses openLibrary() from C3, issueBook() from C2, and readBook() from C1.

Result:

Thus, the program was successfully executed and inheritance among C1, C2 and C3 was verified successfully.